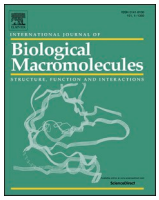
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PreAlgPro: Prediction of allergenic proteins with pre-trained protein language model and efficient neutral network

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ABSTRACT

Allergy is a prevalent phenomenon, involving allergens such as nuts and milk. Avoiding exposure to allergens is the most effective preventive measure against allergic reactions. However, current homology-based methods for identifying allergenic proteins encounter challenges when dealing with non-homologous data. Traditional machine learning approaches rely on manually extracted features, which lack important protein functional characteristics, including evolutionary information. Consequently, there is still considerable room for improvement in existing methods. In this study, we present PreAlgPro, a method for identifying allergenic proteins based on pre-trained protein language models and deep learning techniques. Specifically, we employed the ProtT5 model to extract protein embedding features, replacing the manual feature extraction step. Furthermore, we devised an Attention-CNN neural network architecture to identify potential features that contribute to the classification of allergenic proteins. The performance of our model was evaluated on four independent test sets, and the experimental results demonstrate that PreAlgPro surpasses existing state-of-the-art methods. Additionally, we collected allergenic protein samples to validate the robustness of the model and conducted an analysis of model interpretability.

1. Introduction

Allergies are characterized by an aberrant immune system response to foreign substances known as allergens. Type I allergic reactions, which are triggered by common allergens such as certain food items (such as nuts, fish, milk, and eggs), grass, house dust mite excrement, latex, certain medications, pollen, and insect venoms [1–4]. Upon initial exposure to allergens, hypersensitivity reactions occur, leading to effector responses upon subsequent encounters [5]. These responses involve the production of immunoglobulin E (IgE) to counteract the allergens. Upon re-exposure, sensitized cells coated with IgE antibodies release various mediators, including histamine and leukotrienes. These mediators can induce allergic reactions and potentially lead to anaphylactic shock or even fatal outcomes [6]. Therefore, the most effective approach to prevent allergic reactions is to avoid contact with allergens. However, there are still numerous undiscovered potential allergenic proteins. Hence, it is crucial to accurately identify allergenic proteins, especially when making dietary selections or administering medications, among other scenarios [7].

In the post-genomic era, handling the exponential growth of protein data presents new challenges for conventional experimental techniques [8,9]. These methods for identifying allergenic proteins often require significant time, labor, and expertise from researchers [10]. Consequently, there is an urgent demand for computational approaches that can effectively annotate allergenic proteins. To address this critical need, the Food and Agriculture Organization of the United Nations (FAO) and the World Health Organization (WHO) have developed comprehensive guidelines [11]. These guidelines proposed two primary methods for allergenic protein prediction. The first method classifies a protein as a potential allergenic protein if it contains a stretch of 6 to 8 consecutive amino acids with identical sequence patterns found in known allergens. The second method evaluates the probability of a protein being an allergen based on its overall sequence similarity to known allergenic proteins. Specifically, if a protein exhibits more than 35 % cumulative sequence similarity within an 80-amino acid window to known allergens, it is classified as an allergen [11]. These two methods employ the concept of homology and utilize sequence alignment techniques to identify allergenic proteins. Implementing these computational strategies aims to expedite the identification of allergenic proteins, thereby fostering progress in diverse domains such as food

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production, therapeutic drug development, and biopharmaceutical research. However, these methods heavily depend on homologous comparison, which presents challenges in identifying non-homologous allergens. As a result, this approach can cause false identifications and significant resource allocation [12].

In recent years, machine learning and deep learning methodologies have emerged as primary computational techniques in diverse protein classification tasks. These methodologies utilize protein descriptors [13] as input, which is acquired through computational tools to capture the physicochemical and compositional characteristics of protein sequences [14]. To date, numerous exemplary methods have been proposed for the identification of allergenic proteins, including AlgPred 2.0 [15], AllerCatPro 2.0 [16], Allerdictor [17], AllergenFP [18], AllerTop v.2 [19], Quantitative Structure-Activity Relationship (QSAR) [20], DeepAlgPro [21], ProAll-D [22], SEP-AlgPro [23] and so on. These methods employ a variety of traditional machine learning and deep learning algorithms, such as Random Forest (RF), K Nearest Neighbors (KNN), eXtreme Gradient Boosting (XGBoost), Support Vector Machine (SVM), and Convolutional Neutral Network (CNN). For instance, Sharma et al. introduced AlgPred 2.0, which took into account protein sequence similarity and integrated compositional features into a machine learning classifier to improve prediction accuracy [15]. AllerCatPro 2.0 applied amino acid sequence and 3D structure similarity to forecast the allergenicity of proteins [16]. Allerdictor employed the k-mer algorithm to extract amino acid sequence information and subsequently wielded the linear SVM for accurate protein allergenicity prediction [17]. The AllergenFP tool converted protein sequences into fixed-length vectors by using five amino acid E-descriptors together with auto-covariance and cross-covariance transformation. Additionally, the Tanimoto coefficient was utilized to compare fingerprints for identifying allergenic proteins [18]. AllerTop v.2 was designed to enhance the performance of AllergenFP based on the same protein descriptors by applying the KNN classifier instead of the Tanimoto coefficient [19]. Yu et al. proposed the QSAR model to predict the binding capabilities of IgE to unknown epitopes by building a multiple regression model combined with the selected chemical descriptors [20]. ProAll-D adopted the long short term memory (LSTM) model to overcome the limitations of traditional machine learning approaches and achieved superior performance on the same benchmark datasets with AllerTop v.2 [22]. DeepAlgPro, proposed by He et al., utilized the one-hot encoding to represent protein sequences and leveraged the CNN with the multi-head self-attention mechanism for the large-scale identification of allergenic proteins [21].

Most existing methods depend on handcrafted amino acid descriptors as the input of deep learning or machine learning techniques. However, the selection of useful features often proves to be laborious and inefficient. Fortunately, the advancement of Natural Language Processing (NLP) technology has introduced pre-trained language models (PLMs) with widespread applications across various domains [24]. In the field of bioinformatics, the emergence of pre-trained protein language models (PPLMs) has made significant contributions to various protein classification tasks [14,25,26]. Through learning from extensive protein corpora, PPLMs can effectively capture valuable biological features within protein sequences. Given the powerful performance of PPLMs, Basith et al. proposed SEP-AlgPro [23], which selects 10 feature descriptors from both PPLMs embedded and traditional features, and combines 10 classifiers to form 70 benchmark models. The ensemble learning approach is utilized for predicting allergenic proteins.

Although SEP-AlgPro is an effective method for predicting allergenic proteins, it integrates 70 benchmark models through ensemble learning, which not only requires high computational requirements but also is a cumbersome process. Therefore, there is an urgent need for a better, more efficient, and concise method that can maximize the advantages of PPLM and better adapt to large-scale allergenic protein prediction tasks. In this study, we introduced a novel approach named PreAlgPro, which employed PPLM and the deep learning framework to achieve efficient identification of allergenic proteins. We use the ProtT5 [27] model to extract protein embedding feature representations from protein sequences. Since the extracted embedded features are universal features representing protein sequences, we designed an Attention-CNN module to perform more detailed feature extraction on the embedded feature representation, capturing potential features related to allergenic protein classification. Finally, we trained the proposed model using a 5-fold cross validation approach. In addition to validation on the benchmark dataset, we also created three independent test sets from publicly available datasets for performance generalization testing. In addition, we have collected the latest released allergenic protein data and conducted more comprehensive performance testing. The experimental results indicate that the PreAlgPro surpasses existing leading methods for predicting allergenic proteins. To increase the interpretability of the model, we conducted interpretability analysis using t-Distributed Stochastic Neighbor Embedding (t-SNE) [28] and Shapely Additive explanation (SHAP) tools.

2. Materials and methods

In this section, we introduce the construction of the benchmark datasets, and the proposed PreAlgPro method for predicting allergenic proteins is described in detail. The overall architecture of PreAlgPro is illustrated in Fig. 1. Firstly, the input proteins are encoded by the ProtT5. Subsequently, the Attention-CNN module incorporates the multi-head attention mechanism and convolutional natural network with residual connection to effectively feature extraction. Finally, the features are sent to a classifier layer for final prediction.

2.1. Benchmark datasets

Constructing a robust and efficient classification model heavily relies on a high-quality benchmark dataset. In this study, we utilized the benchmark dataset adopted in previous studies [21] to ensure a fair and comprehensive comparison with existing methods. The benchmark dataset consists of both allergenic protein and non-allergenic protein sequences. The allergenic protein sequences were collected from the Structural Database of Allergenic Proteins (SDAP) [29], COMPARE [30], WHO/IUIS Allergen Nomenclature Sub-Committee [31], AllergenOnline [32], and UniProt [33]. Additionally, it collected 2427 allergenic proteins and 10,075 non-allergenic proteins from the AllerTop v.2 [19] and AlgPred 2.0 [15]. To eliminate duplicate protein sequences obtained from the data integration process, the CD-HIT [34] program was employed. A total of 17,071 duplicate allergen sequences were removed using CD-HIT, employing a sequence identity threshold of 1 and an alignment coverage of 0.8 for the shorter sequence. The benchmark dataset was refined by excluding 37 protein sequences exceeding a length of 1000. Furthermore, non-allergenic protein sequences exhibiting a similarity of over 40 % were filtered out using CD-HIT. Ultimately, the training dataset includes 2840 allergenic proteins and 2840 non-allergenic proteins, which was used to perform the 5-fold cross-validation. The independent dataset contains 710 allergenic proteins and 710 non-allergenic proteins, which were applied to examine our model and make comparisons with other existing tools.

2.2. Protein sequence feature representation

Proteins are composed of 20 common amino acid residues that can be represented as strings of letters, similar to natural language. In recent years, the application of NLP models has shown promise in leveraging deep-level representations of amino acid sequences. These models, pre-trained on large datasets containing tens or even hundreds of millions of protein data, can capture positional information, reflecting the evolutionary information of amino acids within protein sequences. Considering the advantage of PPLMs, we introduce ProtT5-XL-UniRef50 (denote as ProtT5), a T5-based self-supervised learning model trained on the Big Fantastic Database (BFD) [35] and fine-tuned on the UniRef50

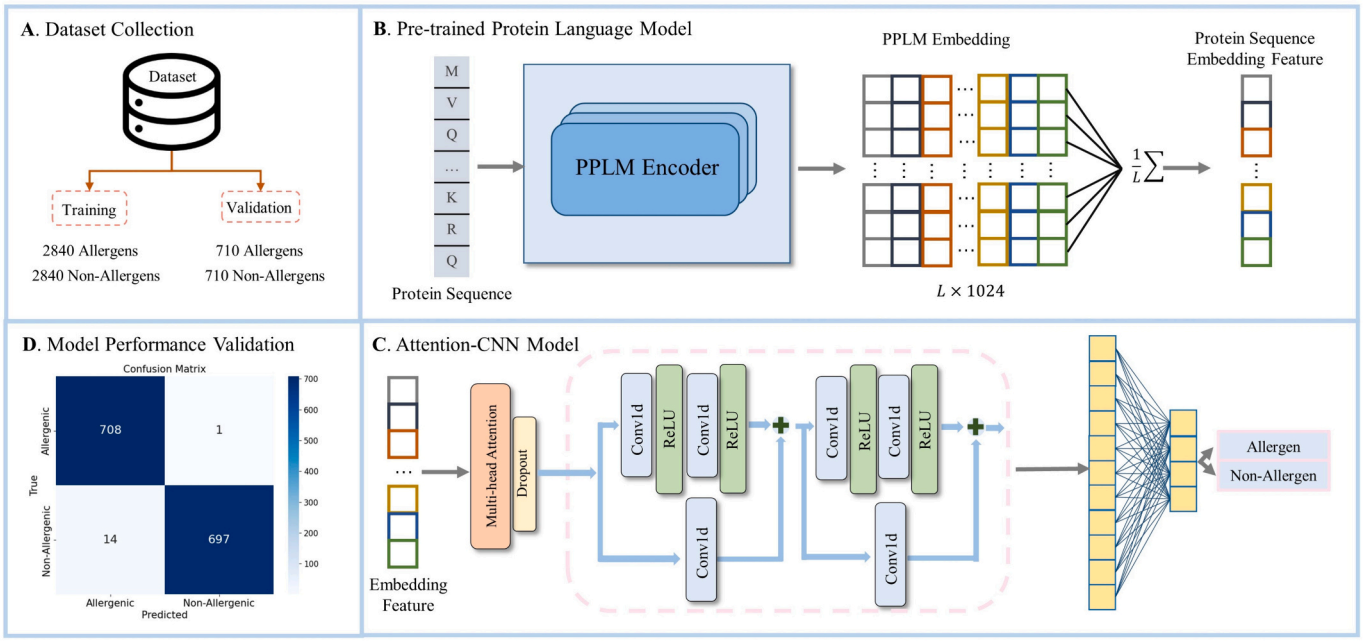


Fig. 1. Overview of the proposed PreAlgPro.

database [36], to capture the contextual information of protein sequences. At first, to prepare the protein sequences for modeling, ambiguous amino acids ('B', 'J', 'O', 'U' and 'Z') are uniformly mapped to 'X', Each protein sequence is treated as a "sentence", where adjacent amino acids are separated by an additional space to indicate a word boundary. Then, each residue is characterized as a 1024-dimemsional vectorized representation by means of the last hidden layer of the encoder in ProtT5. As a result, we can obtain a low-dimensional representation, $X_{protein} \in \Re^{L\times 1024}$, for the protein sequence with L. In addition, due to the different length of each sequence, we perform an average operation on the vectorized representation in the L dimension. Therefore, for each protein sequence, we obtained an embedded feature representation with a length of 1024.

2.3. Architecture of Attention-CNN module

With the emergence of the highly successful Transformer, the self-attention mechanism has found widespread application across various domains. Here, the multi-head attention mechanism is employed in Attention-CNN module to distinguish the contribution of different features. By dividing the embedded feature representations into multiple heads, each head is responsible for learning a distinct subset of crucial information, enabling a more comprehensive understanding of the intricate properties of proteins. The embedded feature representation $F \in \Re^{1024\times 1}$ is initially projected into Query (Q), Key (K) and Value (V) as follows:

$$\begin{cases} Q_i = W_i^q \times F \\ K_i = W_i^k \times F, i = 1, 2, \ldots, H \\ V_i = W_i^v \times F \end{cases} \tag{1}$$

where W_i^q, W_i^q, W_i^q denote the learned weight matrices of Q_i, K_i, V_i, respectively, and H is the number of heads. The the attention scores within each head are calculated and concatenated to generate the final attention representation. The attention function is defined as follows:

$$\mathrm{Head}_i = \mathrm{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i, \mathrm{i} = 1, 2, \ldots, \mathrm{H} \tag{2}$$

$$\mathrm{Attention} = \mathrm{Concat}(\mathrm{Head}_1, \mathrm{Head}_2, \cdots, \mathrm{Head}_h) W^O \tag{3}$$

where W^O is a trainable weighted matrix. In this study, we employed H = 4 to form the parallel attention layers.

Subsequently, the CNN layer was applied to process the output of the multi-head attention mechanism, which consisted of two convolutional blocks. Specifically, each block comprises three 1D convolutional layers and two ReLU activation function. The first convolution block was formulated as follows:

$$\mathrm{CNNLay}_1 = \mathrm{Conv1D}\left(\mathrm{ReLU}\left(\mathrm{Conv1D}\left(F, \frac{F}{2}, 3\right)\right), \frac{F}{2}, 3\right), \tag{4}$$

$$\mathrm{CNNLay}_2 = \mathrm{ReLU}\left(\mathrm{Conv1D}\left(F, \frac{F}{2}, 3\right)\right), \tag{5}$$

$$\mathrm{CNN}_{\mathrm{Block1}} = \mathrm{CNNLay}_1 \bigoplus \mathrm{CNNLay}_2, \tag{6}$$

where the notation $\mathrm{Conv1D}\left(F, \frac{F}{2}, 3\right)$ indicated that the input dimension of the convolution was F, and the output dimension was $\frac{F}{2}$, with a kernel size of 3. CNN_Block1 was the output feature of the first CNN block where "$\bigoplus$" was the concatenation operator. The output of the first convolutional block served as the input to the second convolutional block. While two convolutional blocks shared the same definition of structuring, their dimensions of input and output were not consistent. In this study, $F = 1024$, and the specific parameter changes are depicted in Fig. 2.

2.4. Classification module

As illustrated in Fig. 2, the final classification in the study employed three fully connected layers following the Attention-CNN module. These three fully connected layers, with neuron counts of 128, 64, and 1 respectively, were sequentially utilized to acquire more advanced feature representations. It is important to note that the ReLU activation function and a dropout layer with a dropout rate of 0.2 were applied after each fully connected layer, except for the final layer. Ultimately, the sigmoid activation function was employed to produce the

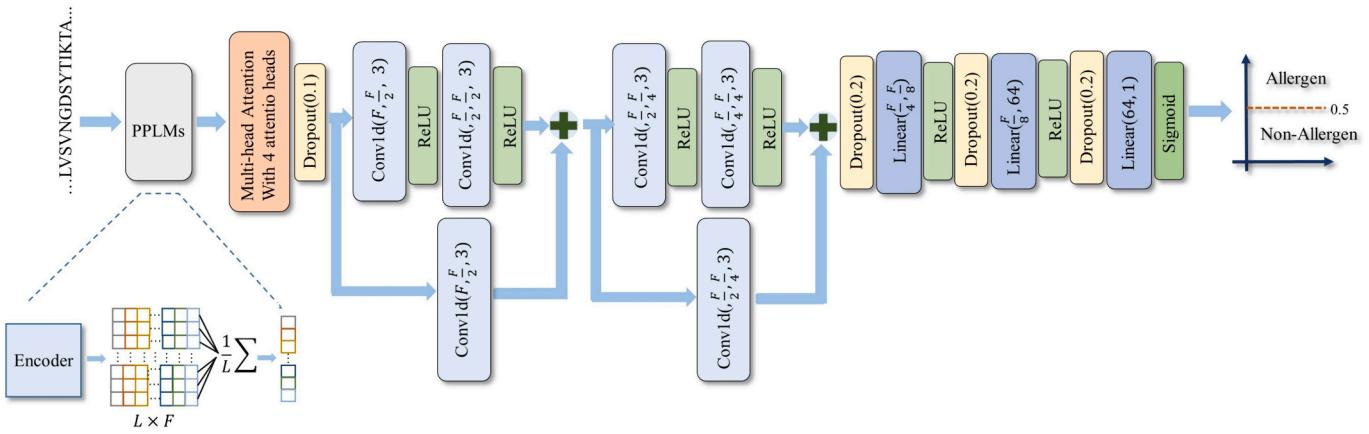


Fig. 2. Overview of the Attention-CNN module and the classification module.

corresponding class probabilities.

2.5. Implementation details

We use the BCELoss as the training loss of PreAlgPro:

$$\text{BCELoss} = -\frac{1}{N}\sum_{i=1}^{N} y_i \bullet log(p(y_i)) + (1-y_i) \bullet log(1-p(y_i)) \quad (7)$$

where y_i denotes the confidence score for the predicted allergenic proteins.

To minimize the loss and optimize the hyperparameters of the PreAlgPro model, the Adam optimization algorithm was employed with a learning rate of 0.00001 and a batch size of 32. Additionally, a 5-fold cross-validation approach was utilized during the training of the model.

2.6. Performance evaluation

To comprehensively evaluate the effectiveness of the proposed method, we performed both the 5-fold cross-validation and the independent test. The four standard evaluation metrics were reported, including Accuracy, Precision, Recall, and F1-score, which are defined as follows:

$$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}}, \quad (8)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad (9)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}, \quad (10)$$

$$\text{F1}-\text{score} = \frac{2\text{TP}}{2\text{TP}+\text{FP}+\text{FN}}, \quad (11)$$

where TP (true positive) and TN (true negative) denote the number of correctly predicted allergenic and non-allergenic proteins while FP (false positive) and FN (false negative) indicate the number of incorrectly predicted allergenic and non-allergenic proteins. In addition, we generated the receiver operating characteristic (ROC) curves and the Precision-Recall (PR) curves. The area under the ROC curve (AUC) and the area under the PR curve (AUPR) were computed as a useful evaluation metric.

3. Results and discussion

3.1. Ablation study

3.1.1. Analysis of the pretrained protein language model

To validate the effectiveness of the ProtT5 protein feature representation, we compared it with four others widely used PPLMs: Bepler [37], PLUS [38], ESM-1b [39], and ESM-2 [40]. These models were evaluated using 5-fold cross-validation on the training dataset, and the introduction of each model are shown in Text S1, and the results are presented in Fig. 3.

Our results indicate that PreAlgPro-ProtT5 outperformed the other models in the 5-fold cross-validation on the training set. This suggests that ProtT5 is more adept at extracting valuable semantic information from protein sequences compared to the alternative models. This advantage can be attributed to ProtT5's utilization of the Transformer encoder-decoder architecture, which has demonstrated strong performance in sequence translation tasks. Furthermore, unlike the ESM series, which also relies on the Transformer architecture, ProtT5 was not only pre-trained on the BDP database but also fine-tuned on the UniRef50 database, resulting in enhanced performance. Among the five models, the Bepler model, based on the BiLSTM architecture, exhibited the lowest performance. This can be attributed to the lower dimensionality of the embedded features it generates. In contrast to the 1024-dimensional embeddings produced by ProtT5, Bepler generates only 121-dimensional representations, thus containing less informative content. Considering the significant advantages demonstrated by ProtT5, we selected it for use in PreAlgPro to generate embedded features from protein sequences.

To further elucidate the reasons behind of ProtT5's outstanding performance in classifying allergenic proteins, we employed the t-SNE tool for embedded feature visualization. In addition to the five PPLMs discussed above, we also encoded the protein sequences using one-hot encoding. The t-SNE visualization results are shown in Fig. 1D. These results indicate that the one-hot feature representation does not effectively capture the differences between allergen and non-allergen proteins. When using the PPLMs, the differentiation between positive and negative samples is markedly improved, consistent with the experimental results. Among these models, the Bepler embedded features showed minimal positive and negative sample aggregation. In contrast, the PLUS, ESM-1b, ESM-2, and ProtT5 models exhibited significant aggregation of features. However, there is a notable overlap between positive and negative samples in the ESM series, which causes performance degradation. The t-SNE tools visualization results further illustrate that the ProtT5 model can effectively capture subtle differences between allergenic and non-allergenic proteins, generate robust feature representations for proteins.

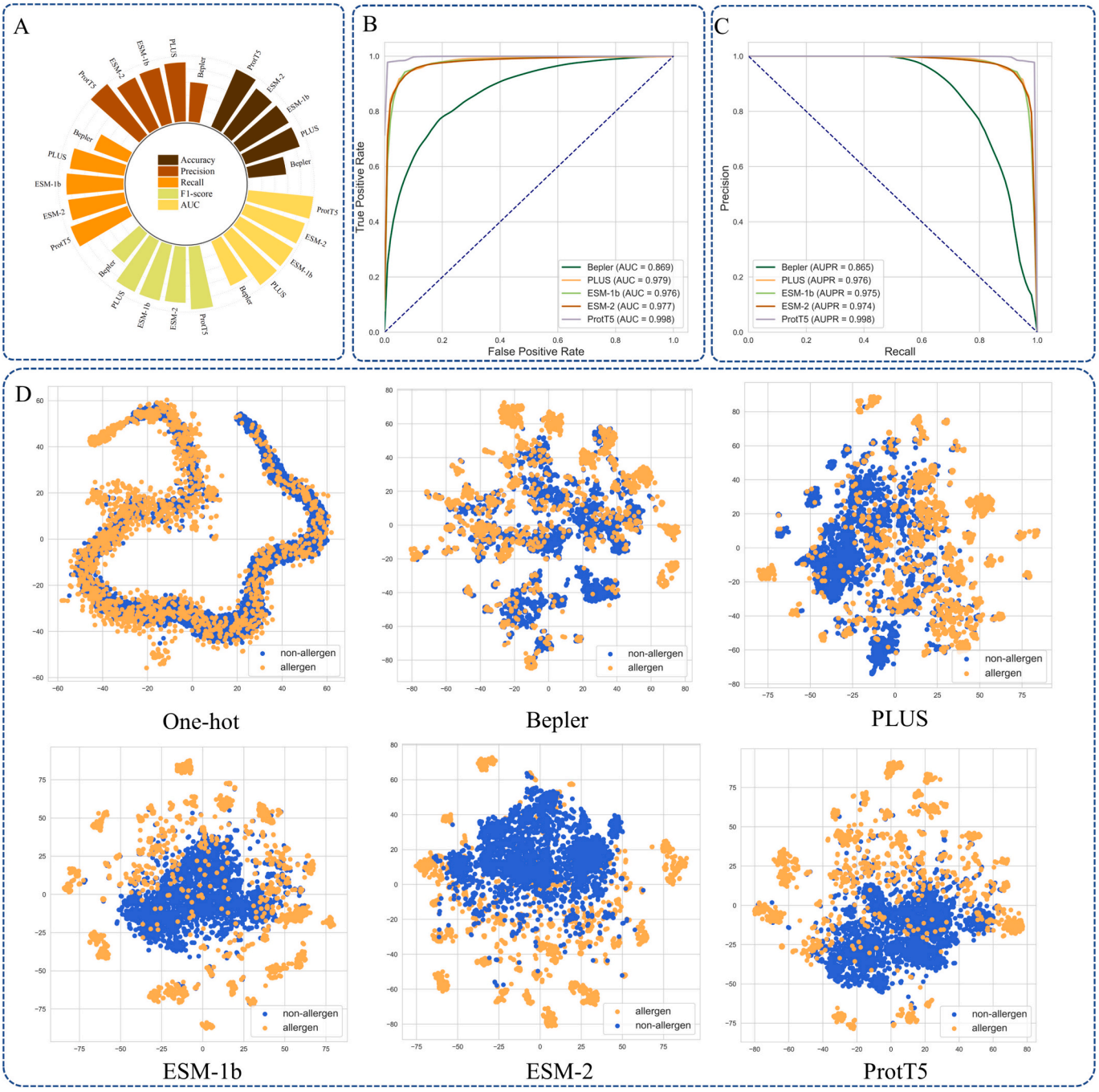


Fig. 3. Performance comparison of PreAlgPro using different pPPLMs on the training datasets. Therein, A show the experimental results in terms of Accuracy, Precision, Recall, F1-score and AUC. B and C are the ROC and PR curves for various models. D is the embedding visualization result generated by t-SNE.

3.1.2. Exploration of the optimal architecture of the PreAlgPro

In this study, we developed PreAlgPro, a model that leverage PPLMs for protein embedding feature extraction. Additionally, we incorporated an Attention-CNN module to further extract fine-grained features pertinent to allergenic protein classification from general features. To evaluate the efficacy of this strategy and the contributions of each module, we designed several variant models.

The benchmark model, M0, utilized one-hot encoding to convert protein sequences into encoded features, followed by fully connected layers for classification. Model M1 employed ProtT5 to extract embedded features from protein sequences, subsequently classified using fully connected layers. Building on M1, we introduced a multi-head attention mechanism, resulting in the variant model M2. Finally, we enhanced M2 by integrating the convolutional module designed in this study, transforming the neural network module into an Attention-CNN module, which constitutes the proposed PreAlgPro. The results presented in Table 1 and the ROC and PR curve results are shown in Fig. 4.

Table 1
Performance comparison of ablation experiment.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score	AUC
M0	0.701	0.741	0.643	0.688	0.787
M1	0.911	0.892	0.936	0.913	0.963
M2	0.963	0.965	0.961	0.963	0.990
PreAlgPro	0.979	0.996	0.963	0.979	0.998

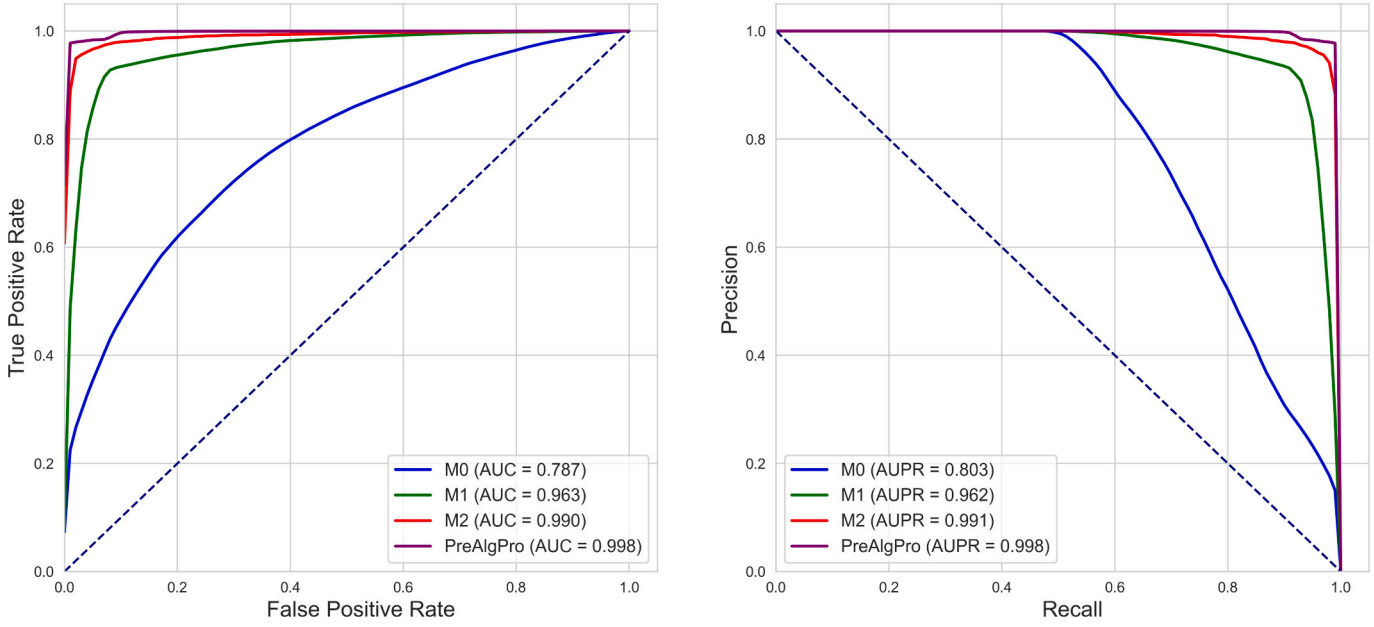


Fig. 4. The ROC and PR curves of ablation experiment.

As shown in Table 2, the benchmark model M0 achieved an accuracy of 0.701, a precision of 0.741, a recall of 0.643, an F1-score of 0.688, and an AUC of 0.787. When our protein sequence feature representation method was changed from one-hot encoding to ProtT5 model, the performance of the model significantly improved. The accuracy, precision, recall, F1-score, and AUC are improved by 21.01 %, 15.11 %, 29.22 %, 22.5 %, and 17.6 %. This indicates that using PPLM is an effective strategy. By pre-training on large databases, it can capture more evolutionary and structural information about proteins, which is beneficial for distinguishing allergenic proteins. Further, when utilizing the attention mechanism, the accuracy, precision, recall, F1-score, and AUC are improved by 5.16 %, 7.28 %, 2.7 %, 5 %, 2.7 %. Since ProtT5 extracts universal features that characterize proteins, the multi-head attention mechanism can better allocate greater weights to potential features related to allergenic protein classification, thereby enhancing model performance. Additionally, when the CNN module was added, the performance of the model further improved. The PreAlgPro achieved an accuracy of 0.979, a precision of 0.996, a recall of 0.963, an F1-score of 0.979, and an AUC of 0.998. From the final experimental results, it can be observed that the Attention-CNN module maximizes the utilization of the embedded features extracted by ProtT5, resulting in a significant increase in the model's performance and demonstrating the effectiveness of each module in PreAlgPro.

3.2. Comparison with widely used deep learning algorithms

To further validate the effectiveness of the proposed Attention-CNN module in the allergenic protein identification tasks, we compared its performance with several widely used deep learning algorithms. Specifically, we evaluated six neural networks: Transformer, CNN, LSTM, BiLSTM, GRU, and BiGRU. In the parameter configuration for these models, the multi-head attention in the Transformer included 4 attention heads with a total of 4 layer of Transformer encoders. For the CNN, we designed three layers of convolutional layers with kernel sizes of 3, 5, and 7, allowing for a rich receptive field. All other models used default parameters. The experimental results are presented in Table 2, with the ROC and PR curves are shown in Fig. 5.

Table 2
Performance comparison between PreAlgPro and six widely used deep learning algorithms.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score	AUC
Transformer	0.931	0.933	0.920	0.931	0.974
CNN	0.952	0.948	0.957	0.952	0.985
LSTM	0.920	0.919	0.922	0.920	0.965
BiLSTM	0.928	0.933	0.923	0.928	0.975
GRU	0.926	0.919	0.935	0.927	0.969
BiGRU	0.941	0.936	0.945	0.941	0.981
PreAlgPro	0.979	0.996	0.963	0.979	0.998

For the results, it is evident that among the six deep learning algorithms compared, the CNN network achieved the best performance, with an accuracy of 0.952, a precision of 0.948, a recall of 0.957, an F1-score of 0.952, and an AUC of 0.985. BiGRU and Transformer also performed well, with accuracy of 0.941 and 0.931, precision of 0.936 and 0.933, recall of 0.945 and 0.920, F1-score of 0.941 and 0.931, and AUC of 0.981 and 0.974, respectively. Additionally, LSTM, BiLSTM, and GRU all achieved accuracy exceeding 0.920.

We also compared different deep learning methods on four other PPLMs, and the results are shown in Table S1. From the results, it can be found that in most cases, BiLSTM performs better than LSTM, and BiGRU achieves better performance than GRU. We also found that among the six models compared, CNN networks can achieve good results, but Transformer networks have unstable performance. The designed Attention-CNN network surpasses Transformer, CNN, LSTM, BiLSTM, GRU, and BiGRU in predicting allergenic proteins, potentially due to its capacity to capture both global and local information in sequences through the attention mechanism and convolutional network. This ability enhances feature representation, processes long sequences more efficiently, and provides strong interpretability. In contrast, while CNNs are restricted by the size of the convolutional kernel for capturing global information, Transformers can be limited by the number of encoder layers, potentially leading to a lack of local information. Although LSTM, BiLSTM, GRU, and BiGRU are capable of handling long-range dependencies, they often come with high computational complexity and are prone to issues like gradient vanishing or explosion.

3.3. Performance comparisons of PreAlgPro with existing state-of-the-art methods

To provide a comprehensive assessment of the efficacy of our proposed PreAlgPro, we conducted a comprehensive analysis using the

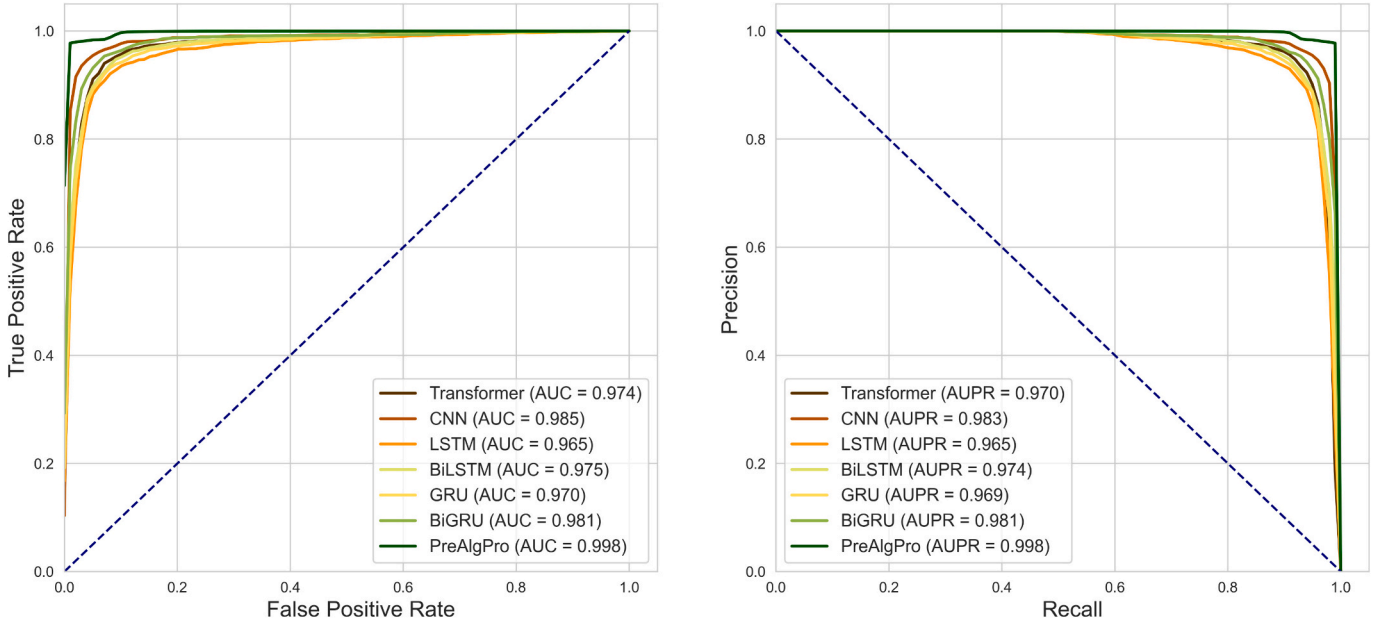


Fig. 5. The ROC and PR curves of PreAlgPro and six widely used deep learning algorithms.

same test set and benchmarked it against several state-of-the-art methods, including ProAll-D, AllergenPF, AllerTop v.2, AllerCatPro 2.0, AlgPred 2.0, DeepAlgPro, and SEP-AlgPro. The evaluation results are summarized in Table 3.

Table 3 presents the performance of eight methods on the independent test set, including accuracy, precision, recall, F1-score, and AUC. PreAlgPro outperforms all other methods across all metrics. DeepAlgPro, a deep learning method that utilizes one-hot encoding for protein sequence feature transformation, achieves excellent performance with an accuracy of 0.916. SEP-AlgPro, a deep learning-based method similar to the proposed approach, utilizes PPLMs for feature extraction. It combines 10 PPLMs and 7 classifiers in an ensemble manner, achieving an accuracy of 0.957 and a precision of 0.975. AllerCatPro 2.0, a structure-based method that utilizes structural information for allergenic protein classification, demonstrates outstanding performance with an accuracy of 0.938 and a recall of 0.944.

In Table 3, it can be observed that SEP-AlgPro and PreAlgPro outperform other methods. Particularly, compared to DeepAlgPro, the embedding features extracted using PPLMs exhibit more potential features relevant to allergenic protein classification. Furthermore, in contrast to traditional machine learning classifiers, deep learning methods show superior performance. Among the methods utilizing handcrafted features, DeepAlgPro demonstrates more competitive results, while PreAlgPro, employing deep learning methods as well, achieves satisfactory outcomes. These comparative results underscore the effectiveness of PreAlgPro in allergenic protein classification tasks. By leveraging ProtT5 to extract embedding features and combining them with a specially designed Attention-CNN network, PreAlgPro enables the sequence-based DeepAlgPro method to adapt to large-scale prediction tasks.

Table 3
Performance comparison of PreAlgPro with the existing state-of-the-art predictors on independent test set.

Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-score	AUC
ProAll-D	0.860	0.868	0.849	0.858	NA
AllergenPF	0.876	0.866	0.890	0.878	NA
AllerTop v.2	0.898	0.912	0.880	0.896	NA
AllerCatPro 2.0	0.938	0.933	0.944	0.938	NA
AlgPred 2.0	0.903	0.851	0.976	0.909	0.984
DeepAlgPro	0.916	0.928	0.903	0.915	0.953
SEP-AlgPro	0.957	0.975	0.976	0.956	0.985
PreAlgPro	0.989	0.999	0.980	0.989	0.998

3.4. Extensive validations of PreAlgPro using three independent test sets

The dataset adopted in this study is a balanced dataset. In reality, the count of non-allergenic proteins significantly outweighs that of allergenic proteins. To thoroughly assess the efficacy of the proposed PreAlgPro on imbalanced datasets, three additional independent test sets were employed for performance validation. These datasets were denoted as TE1, TE2, and TE3, respectively. TE1 was derived from the independent test set of AlgPred 2.0 [15], yielding 171 allergenic proteins and 1810 non-allergenic proteins after removing samples overlapping with the PreAlgPro training set. TE2 was based on an independent test set from NetAllergen [41], which, after excluding overlapping data, included 928 allergenic proteins and 4740 non-allergenic proteins. TE3 combined the training sets of AlgPred 2.0 and NetAllergen, resulting in 550 allergenic proteins and 4769 non-allergenic proteins after duplicate sequences were removed. The details of these test sets are summarized in Table 4. We compared the performance of DeepAlgPro, SEP-AlgPro, and PreAlgPro on these test sets. The results, presented in Table 5, indicate that PreAlgPro consistently outperformed the other methods across all datasets.

On TE1, PreAlgPro achieved an accuracy of 0.988, a precision of 1.000, a recall of 0.860, an F1-score of 0.925, and an AUC of 0.999. Compared to DeepAlgPro, PreAlgPro showed improvements of 6.5 % in accuracy, 46.6 % in precision, 1.2 % in recall, 26.9 % in F1-score, and 5.7 % in AUC. When compared to SEP-AlgPro, PreAlgPro demonstrated increases of 1.4 % in accuracy, 18.9 % in precision, 0.2 % in recall, 6.9 % in F1-score, and 4.1 % in AUC. On TE2, PreAlgPro achieved an accuracy of 0.987, a precision of 0.998, a recall of 0.925, an F1-score of 0.960, and

Table 4
Summarize of three independent test sets.

Datasets	Allergens	Non-allergens
TE1	171	1810
TE2	928	4740
TE3	550	4769

Table 5
A comparative analysis of PreAlgPro with two excellent predictors on three independent test sets.

Datasets	Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score	AUC
TE1	DeepAlgPro	0.923	0.534	0.848	0.656	0.942
	SEP-AlgPro	0.974	0.811	0.906	0.856	0.960
	PreAlgPro	0.988	1.000	0.860	0.925	0.999
TE2	DeepAlgPro	0.879	0.594	0.822	0.677	0.928
	SEP-AlgPro	0.941	0.774	0.901	0.834	0.956
	PreAlgPro	0.987	0.998	0.925	0.960	0.996
TE3	DeepAlgPro	0.906	0.542	0.938	0.674	0.972
	SEP-AlgPro	0.937	0.627	0.958	0.760	0.966
	PreAlgPro	0.977	0.871	0.918	0.894	0.987

an AUC of 0.996. Compared to DeepAlgPro, PreAlgPro's metrics showed improvements of 10.8 % in accuracy, 40.4 % in precision, 10.3 % in recall, 28.3 % in F1-score, and 6.8 % in AUC. Compared to SEP-AlgPro, PreAlgPro showed increases of 4.6 % in accuracy, 22.4 % in precision, 2.4 % in recall, 12.6 % in F1-score, and 4.2 % in AUC. On TE3, PreAlgPro achieved an accuracy of 0.977, a precision of 0.871, a recall of 0.918, an F1-score of 0.894, and an AUC of 0.987. Compared to DeepAlgPro, these metrics improved by 7.1 % in accuracy, 32.9 % in precision, 2.0 % in recall, 32.5 % in F1-score, and 1.5 % in AUC. Compared to SEP-AlgPro, PreAlgPro showed increases of 4.0 % in accuracy, 24.4 % in precision, 1.3 % in recall, 17.6 % in F1-score, and 2.1 % in AUC. The outcomes underscore the efficacy of PreAlgPro, showcasing its competitive edge over established methods across three imbalanced independent test sets. Notably excelling in accuracy, these findings highlight PreAlgPro as a reliable model with exceptional robustness and robust generalization capabilities, adept at tackling the complexities inherent in imbalanced datasets. The combined utilization of ProtT5 and the Attention-CNN network proves instrumental in capturing crucial features closely linked to allergenic proteins.

3.5. Model interpretation of PreAlgPro

To investigate the effectiveness of the PreAlgPro architecture, we extracted intermediate layer outputs from each stage of the model and employed the t-SNE tool to map high-dimensional features to a two-dimensional space. Fig. 6A displays the embedded features of the original ProtT5 on the test set, showing a chaotic distribution. This is because the PPLM extracts a universal feature representation rather than a refined representation to particular type of protein. However, after applying the multi-head attention module, the features' clustering effect becomes apparent. The multi-head attention mechanism effectively extracts potential features related to allergenic protein classification from the general feature representation. Subsequently, convolutional neural networks were utilized to further preserve key classification features, resulting in allergenic and non-allergenic proteins appearing in distinct clusters.

Additionally, we explored the impact of different features on the recognition of allergenic proteins using the SHAP tool. SHAP is a commonly used model interpretability tool, and here we use the DeepExplainer interpreter in SHAP to achieve deep learning model interpretability analysis. Fig. 6B presents a scatter plot of feature density, with each row representing a feature. The horizontal axis represents the SHAP value, and the features are sorted according to the average absolute SHAP value. The left panel shows the 20 features that most influential for allergenic proteins, while the right panel shows the 20 features that most influential for non-allergenic protein identification. Each point represents a sample, with red indicating higher feature values and blue indicating lower feature values. The figure reveals that some features are crucial for identifying both allergenic and non-allergenic proteins, highlighting the presence of effective feature information in the features extracted by the original ProtT5 model and the need for further extraction through advanced neural networks.

Fig. 6C illustrates how features 591, 636, and 589 interact with other features. We observed strong correlations between feature 591 and feature 973, feature 636 and feature 591, and feature 589 and feature 433. For example, when the special 591 value is low, the impact of feature 973 on classification is diminished. This is because ProtT5 includes global context dependencies, and the expression of features is based on their interaction with other features, further confirming the effectiveness of ProtT5.

Moreover, we calculated the single sample feature influence map, as shown in Fig. 6D. We determined the feature influence results of one non-allergenic protein sample and one allergenic protein sample. Among them, red represents positive influence, and blue represents negative influence. The first image of Fig. 6D shows the feature influence map of the non-allergenic protein sample, where most features negatively impact the classification results, while a few positively influence the classification. The model's output for this sample was −0.47. The second image shows the feature influence map of the allergenic protein, which is the opposite of the first image. The model's output for this sample was 1.26. Through these interpretable explanations, we can better and more intuitively understand the reasons behind PreAlgPro's outstanding performance.

3.6. Case study

The ubiquity of modified proteins in daily life underscores the growing significance of identifying potential allergenic proteins. To rigorously assess the robustness of our novel algorithm, PreAlgPro, we conducted a comprehensive evaluation of its ability to detect new allergens using existing tools as benchmarks. We collected the latest protein sequences of allergenic proteins submitted to the WHO/IUIS Allergen Nomenclature Sub-Committee [31] between January 2022 and November 2023. After excluding duplicates of allergenic proteins present in the training and test sets, 75 new samples of allergenic protein sequences were acquired. The results of AllerCatPro 2.0, SEP-AlgPro, and AlgPred 2.0 were obtained from the available website. In addition, the open-source code of DeepAlgPro and the trained model are used to obtain the results of the case analysis. The experimental data and results are summarized in Table S2.

Our findings reveal that PreAlgPro achieved an impressive accuracy rate, correctly predicting 73 out of 75 new allergenic samples. In comparison, AlgPred 2.0, DeepAlgPro, AllerCatPro 2.0, and SEP-AlgPro accurately identified 72, 71, 70, and 68 allergen proteins, respectively. It is noteworthy that two allergenic protein samples derived from opium poppy were overlooked by these methods. Despite these occasional misses, PreAlgPro stands out for its exceptional efficacy in identifying novel allergens, outperforming existing tools. Moreover, SEP-AlgPro, which also utilizes a pre-trained protein language model, showed less than optimal performance. This comparative analysis underscores the superior robustness of PreAlgPro, particularly in its ability to handle entirely new samples with high accuracy. Our results not only highlight the potential of PreAlgPro as a leading tool in allergen prediction but also emphasize its robustness and reliability in the face of emerging allergenic proteins.

4. Conclusion

Accurately predicting allergenic proteins holds significant importance across various domains, including food production and drug development. However, the current method employed for identifying these proteins through web lab experiments are time-consuming, labor-intensive, and expensive. Unfortunately, existing homology-based methods cannot effectively identify non-homologous proteins. In addition, machine learning methods based on experimental manual features lack limited deep level protein information, resulting in low recognition efficiency and unsatisfactory accuracy. Therefore, we propose a novel

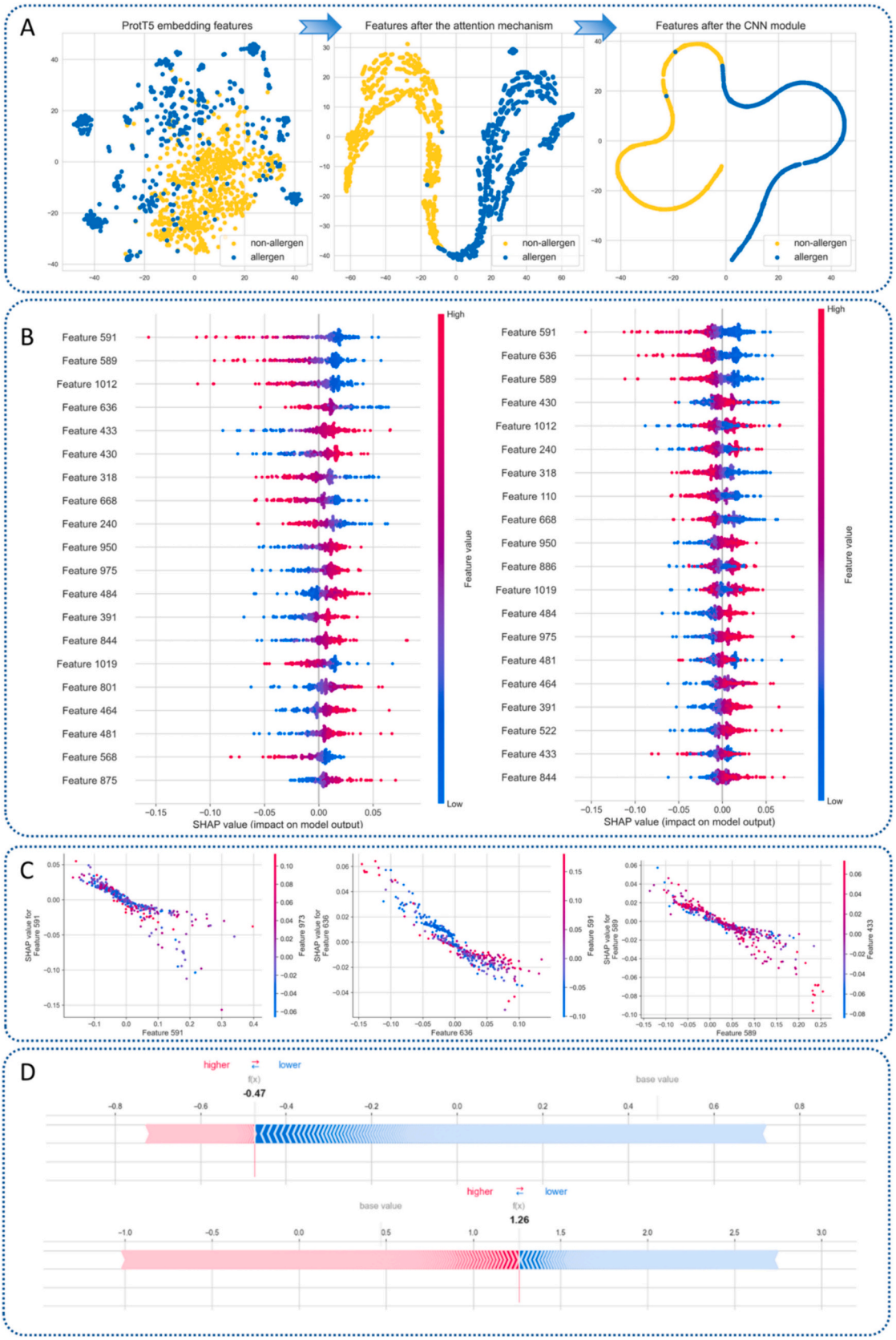


Fig. 6. Presentation of the interpretability experiment results. A is the feature visualization in each stage by t-SNE tool. B is the top 20 features with the greatest impact on predicting allergen proteins and non-allergen proteins. C is the effect of interactions between features. D is the single sample feature influence map of a positive sample and a negative sample.

method for identifying allergenic proteins, PreAlgPro. We use ProtT5 to extract embedded representations from protein sequences, and then use the designed Attention-CNN model to learn potential features related to allergenic protein classification. To validate the performance of the model, we tested it on an independent test set and three additional test sets, and the results showed that PreAlgPro surpassed the existing state-of-the-art methods for identifying allergenic proteins. In addition, the robustness of the model was verified through case analysis. We also conducted interpretability analysis on the model using two interpretability tools. For better academic use, we have also open-source the code and trained models on GitHub.

In the task of functional protein classification, features play a crucial role. Currently, many studies revolve around fusing multi-view features to improve the accuracy of protein classification. Therefore, in our feature research, we will also attempt to fuse multiple embedding features extracted by PPLMs. However, it is worth noting that PPLMs only extract general protein sequence representations, so an efficient neural network needs to be designed according to specific tasks. Finally, to facilitate the use of biological researchers, we will also work on building a comprehensive web server to further enhance the convenience of model usage. We hope our research can provide greater convenience to researchers.

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CRediT authorship contribution statement

Lingrong Zhang: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Taigang Liu:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data and code for this study can be found in a GitHub repository accompanying this manuscript: https://github.com/zlr-zmm/PreAlgPro.

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